# GLM-5: from Vibe Coding to Agentic Engineering

13th AUG

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# Background:

# Vibe Coding vs Agentic Engineering

# Vibe Coding




- Proposed by Andrej Karpathy



Andrej Karpathy
@karpathy

There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

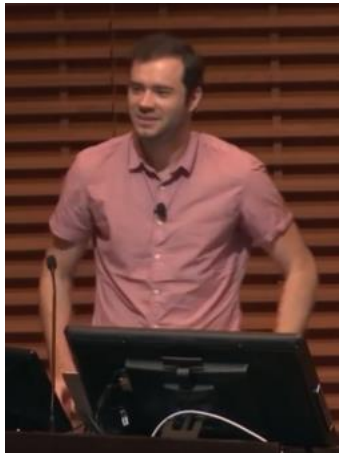
12:17 AM · Feb 3, 2025 · **7.3M** Views

1.5K 3.6K 34K 18K

Elon Musk @elonmusk · Feb 3, 2025
Hmm

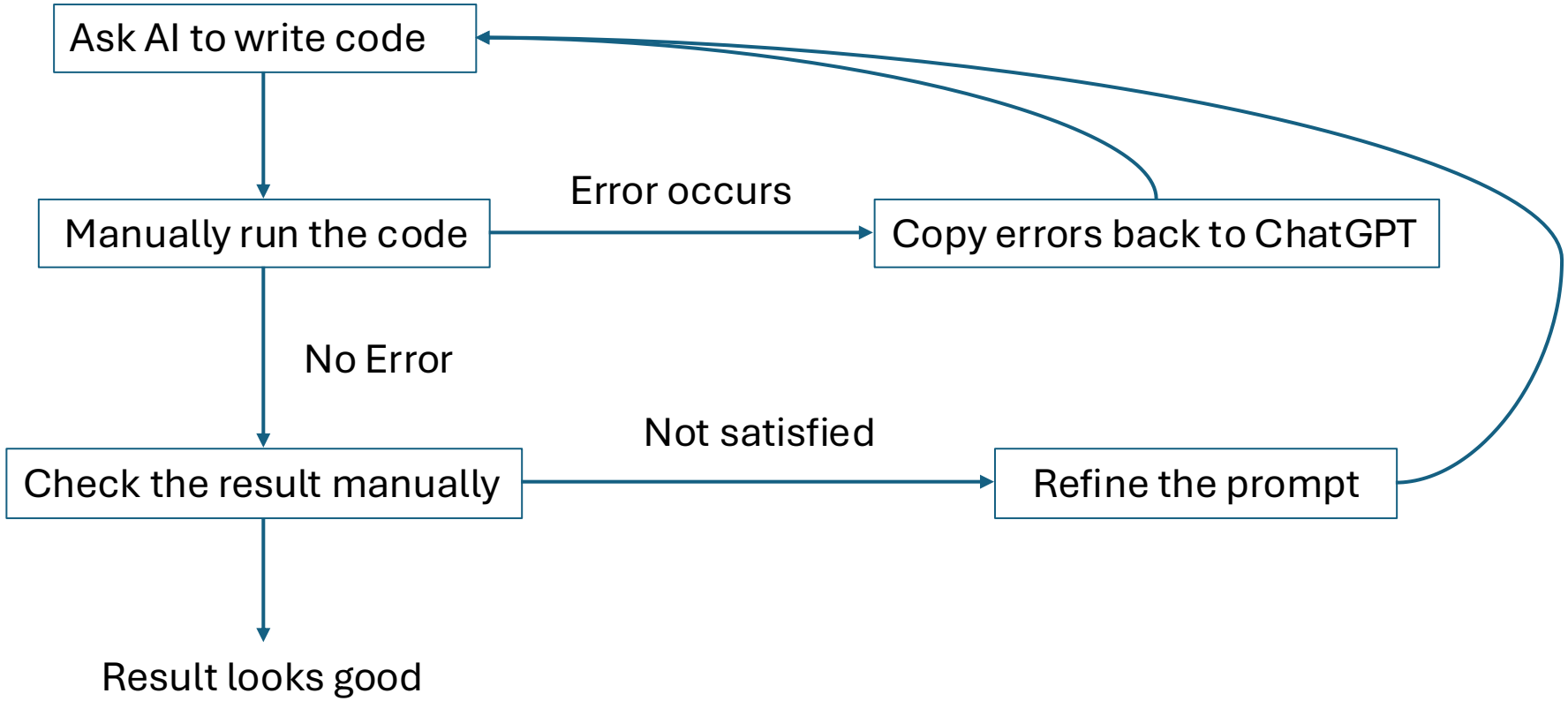
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https://x.com/karpathy/status/1886192184808149383



Simply telling AI what you want, and letting it do the coding.

# Example: Vibe Coding with ChatGPT




❖ In vibe coding, the human stays in the loop

# Vibe Coding

❖ Generate and debug code through conversation

➢ Tools


GPT


Claude


LLaMA


Zhipu

➢ Typical Use Cases of Vibe Coding

Generate a specific function with explicit input and output (code unit);

Create small scripts, e.g., write SQL for search in database;

Paste an error and ask AI to fix

❖ Character: human controls the workflow.

# Agentic Engineering

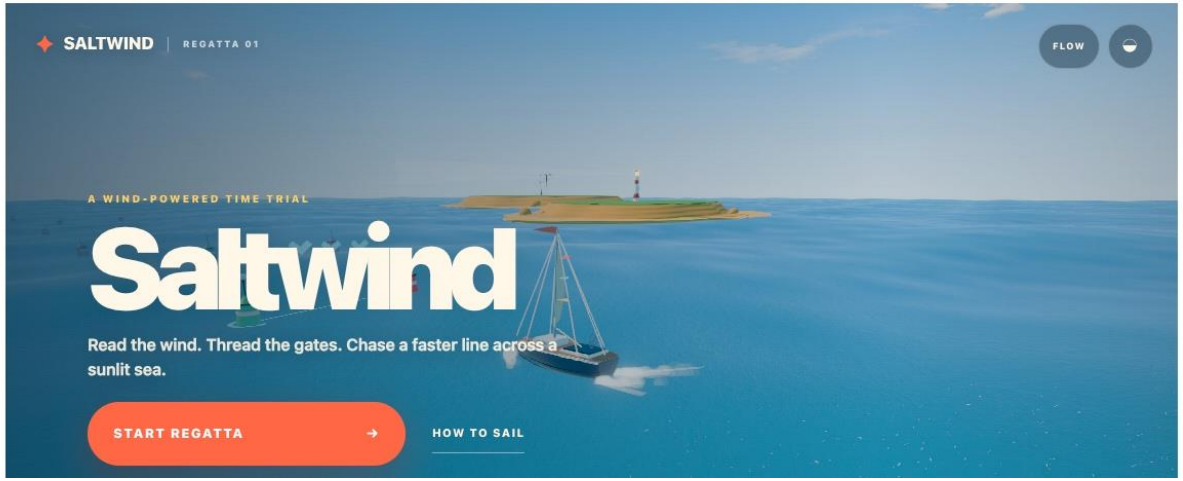
❖ Give AI a task and letting it plan, code, test, and fix problems on its own

**Vibe Coding**: tell AI what code to write;

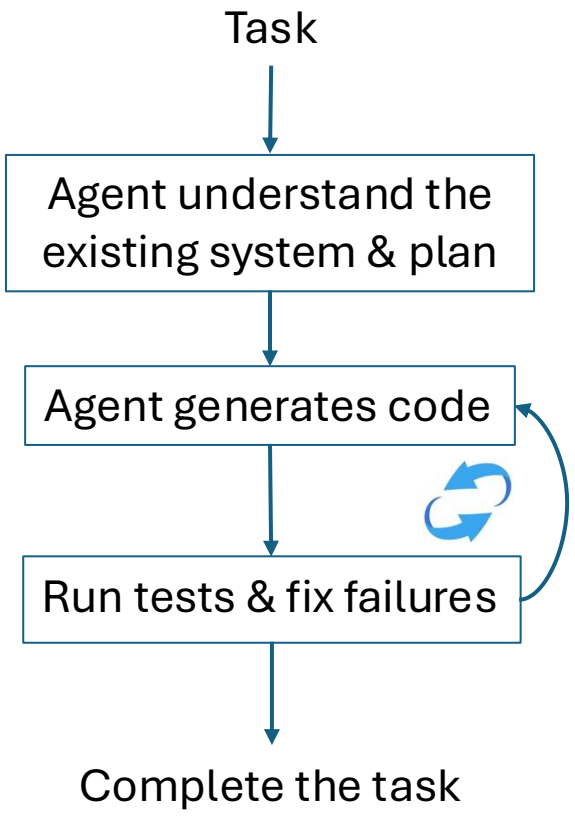
**Agentic Engineering**: just tell AI the task.

➢ Example Task for Agentic Engineering

- ➢ Build a small game
  - Build a sailing game, …
- ➢ Build a webpage
- ➢ Build an application



# Example: Agentic Engineering




- Human defines the goal and rules

> **Task**: *Add a user authentication feature to the existing web application. Users should be able to sign up, log in, and log out. All existing tests must still pass.*

- Agent autonomously plans, implements and tests

# Agentic Engineering Tool



1. **Claude Code:**
   Most commonly used coding agent



2. **OpenAI Codex:**
   handle a wide range of general tasks



3. **GitHub Copilot:**
   From coding assistant to coding agent



4. **Cursor:**
   AI-native code editor built on top of VS Code

# Vibe Coding vs Agentic Engineering

- **Input Example**
  - Vibe Coding: *Write a Python function to calculate the average price in this CSV file.*
  - Agentic Engineering: *Analyze the price in this dataset and build a dashboard to present the results.*

| | Vibe Coding | Agentic Engineering |
|---|---|---|
| **Human Input** | Specific coding requests | High-level task |
| **Planning** | Human decides what to do next | Agent plans autonomously |
| **Debugging** | Human reports errors to AI | Agent detects & fixes errors |
| **Workflow** | Human-driven | Agent-driven |

- **Vibe Coding:**

  Human drives the process, and manually performs step-by-step. (AI writes code for you)

- **Agentic Engineering:**

  Human gives the goal, and the agent drives the process. (AI does the engineering for you)

# How Do We Turn an LLM into a Coding Agent?

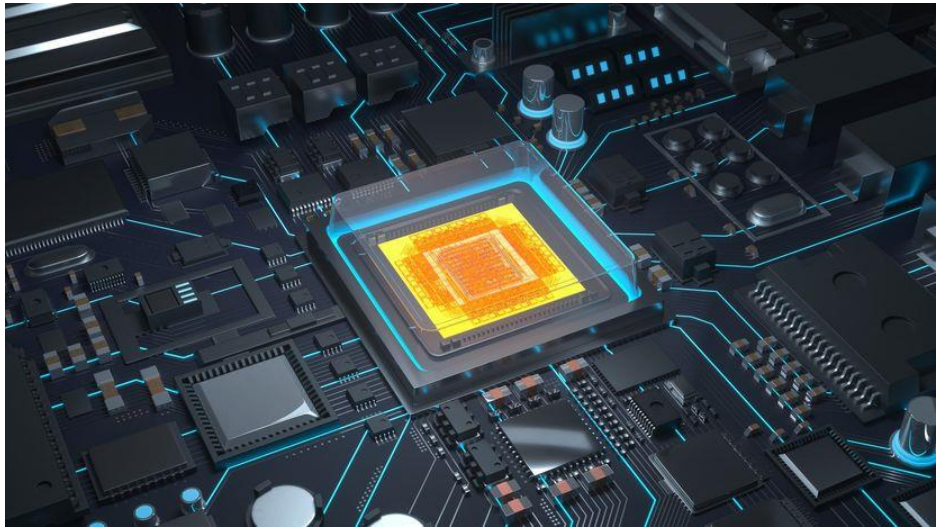
# An LLM Alone Is Not a Coding Agent




**LLM cannot directly:**

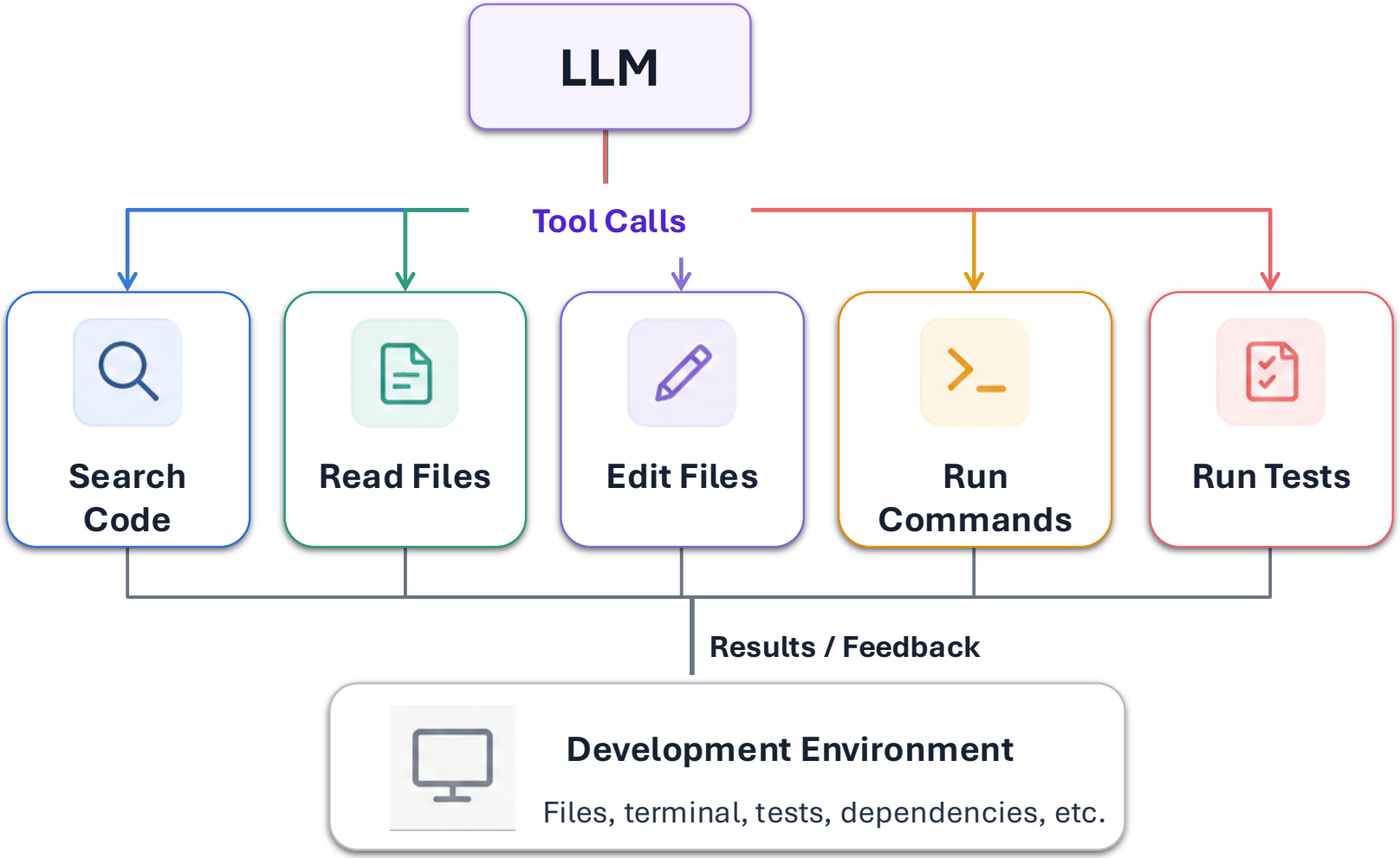
- Read project
- Modify files
- Run code
- Observe running results

**LLM ≈ CPU**
**Agentic System ≈ Computer**



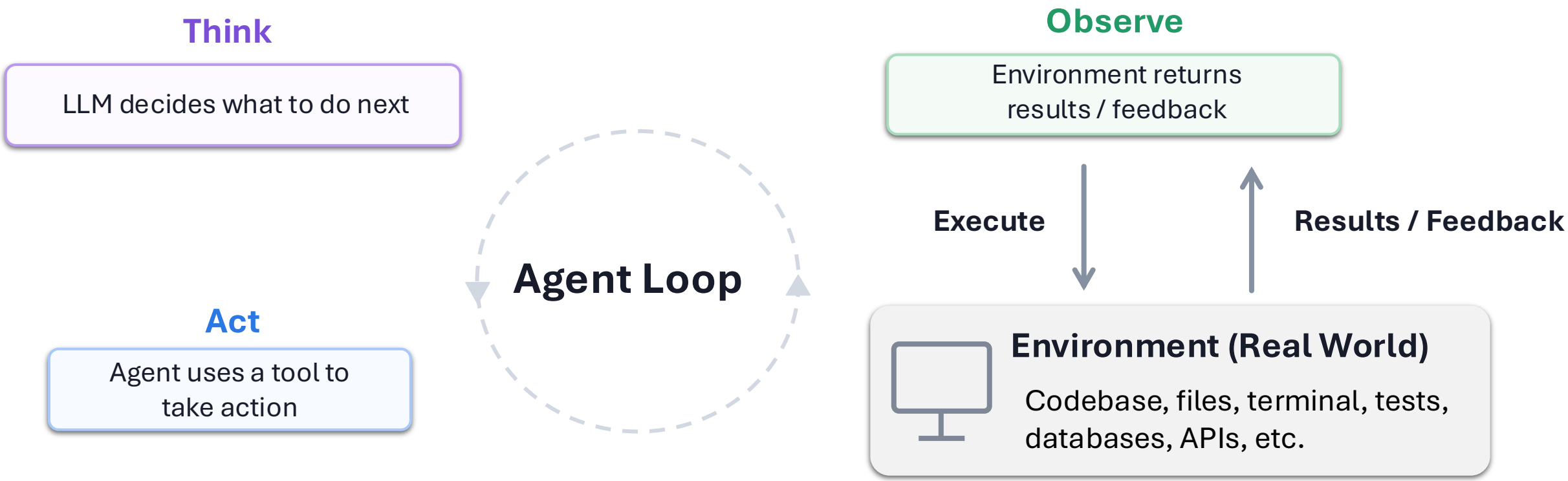
# Give the LLM Tools

Tools give the LLM the ability to interact with the real development environment.



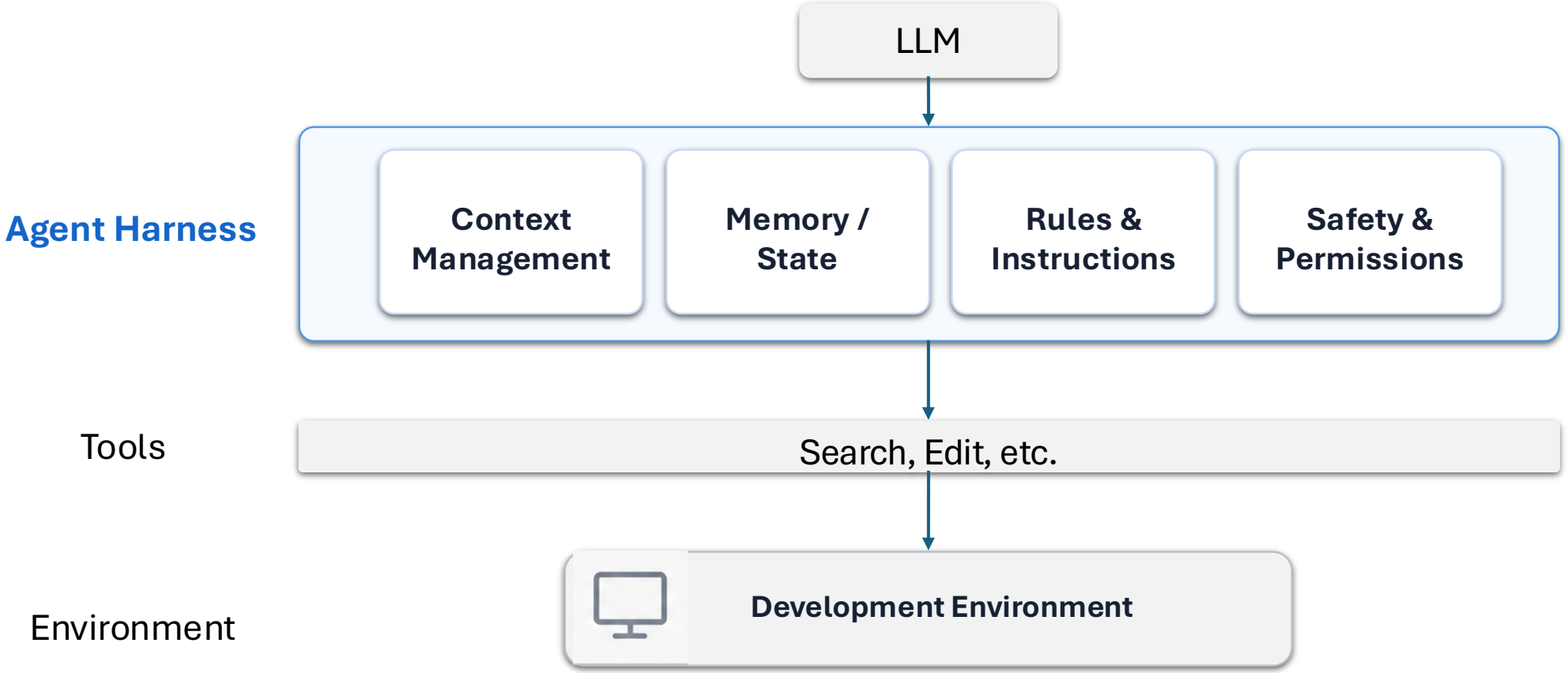
# The Key: Agent Loop

Agent can continuously act, observe, and improve.



# Agent Harness: The System Around the LLM

The harness connects the LLM (brain) with tools, context, and the real environment.

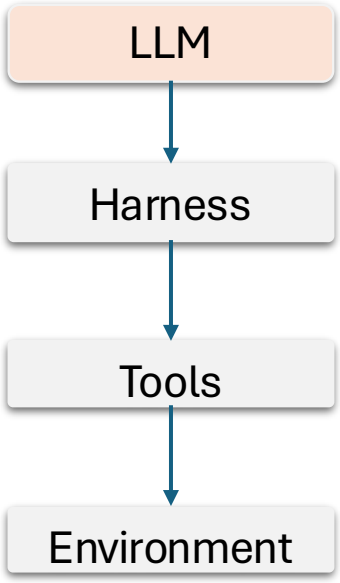


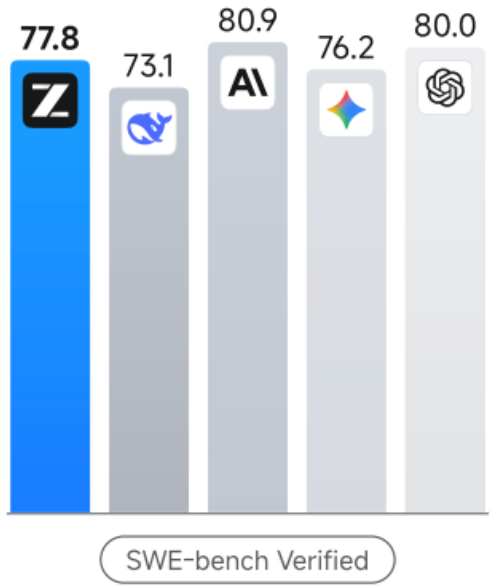

❖ **Coding Agent = LLM + Harness + Tools + Environment**

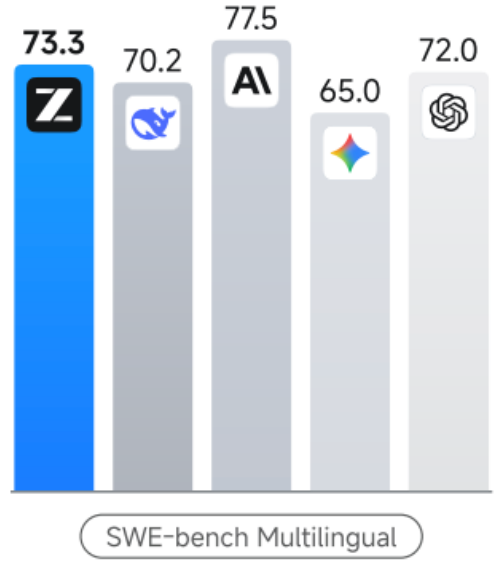
# What Capabilities Does the LLM Need?

# Performance of Different LLMs

❖ Coding Agent = **LLM** + Harness + Tools + Environment



# LLM in Agentic Engineering

➢ **Scale:**

Larger model is more intelligent



➢ **Long Horizon:**

Work for many steps



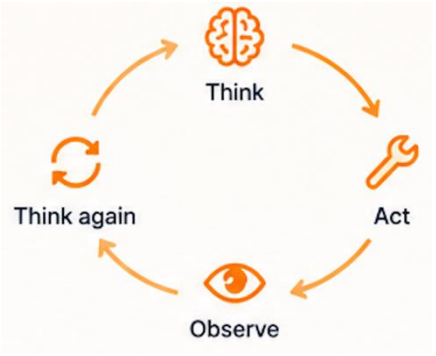
➢ **Tool Use:**

Choose the right tools



➢ **Persistent Reasoning:**

Keep thinking after each action




❖ **GLM-5 is designed around exactly these requirements.**

# Training Data: From Code Generation to Agentic Engineering

➢ Code generation training data

**Prompt**:Write a Python function to check if a string is a palindrome.

LLM

Target (Code)

```
def is_palindrome(s: str) -> bool:
    s = s.lower()
    s = ''.join(c for c in s if c.isalnum())
    return s == s[::-1]
```

➢ Agentic engineering training data

**Task**: Fix the bug in function calculate_ average so that it ignores non-numeric values. The project uses Python.

LLM

Target (Trajectory)

| | | |
|---|---|---|
| 1 | Plan (Reasoning) | I will first locate the file that contains `calculate_average`. |
| 2 | Call Tool (Search) | search_code("calculate_average") |
| 3 | Tool Result | Found in utils/stats.py (line 10) ... |
| 4 | Call Tool (Read File) | read_file("utils/stats.py") |
| 5 | Tool Result | def calculate_average(values): return sum(values) / len(values) ... |
| 6 | Reasoning | The function doesn't filter non-numeric values. I will add a check. |
| 7 | Call Tool (Edit File) | edit_file("utils/stats.py", new_code=...) |
| 8 | Call Tool (Run Tests) | run_tests() |
| 9 | Tool Result (Failed) | Test test_ignore_non_numeric failed ... |
| 10 | Debug (Reasoning) | I missed the case when all values are non-numeric. I will handle that. |
| 11 | Edit Again | edit_file("utils/stats.py", new_code=...) |
| 12 | Run Tests Again | All tests passed ✅ |
| 13 | Finish (PR) | Create pull request with changes |

A complete trajectory

❖ Agentic Models: Learn from the engineering process.

# What’s Next?

1. How does team of GLM 5 build agentic training data?

    Collect high-quality trajectories.

2. How do they train LLMs with these trajectories?

    SFT, reinforcement learning

**Thank you!**

*Happy to take questions.*